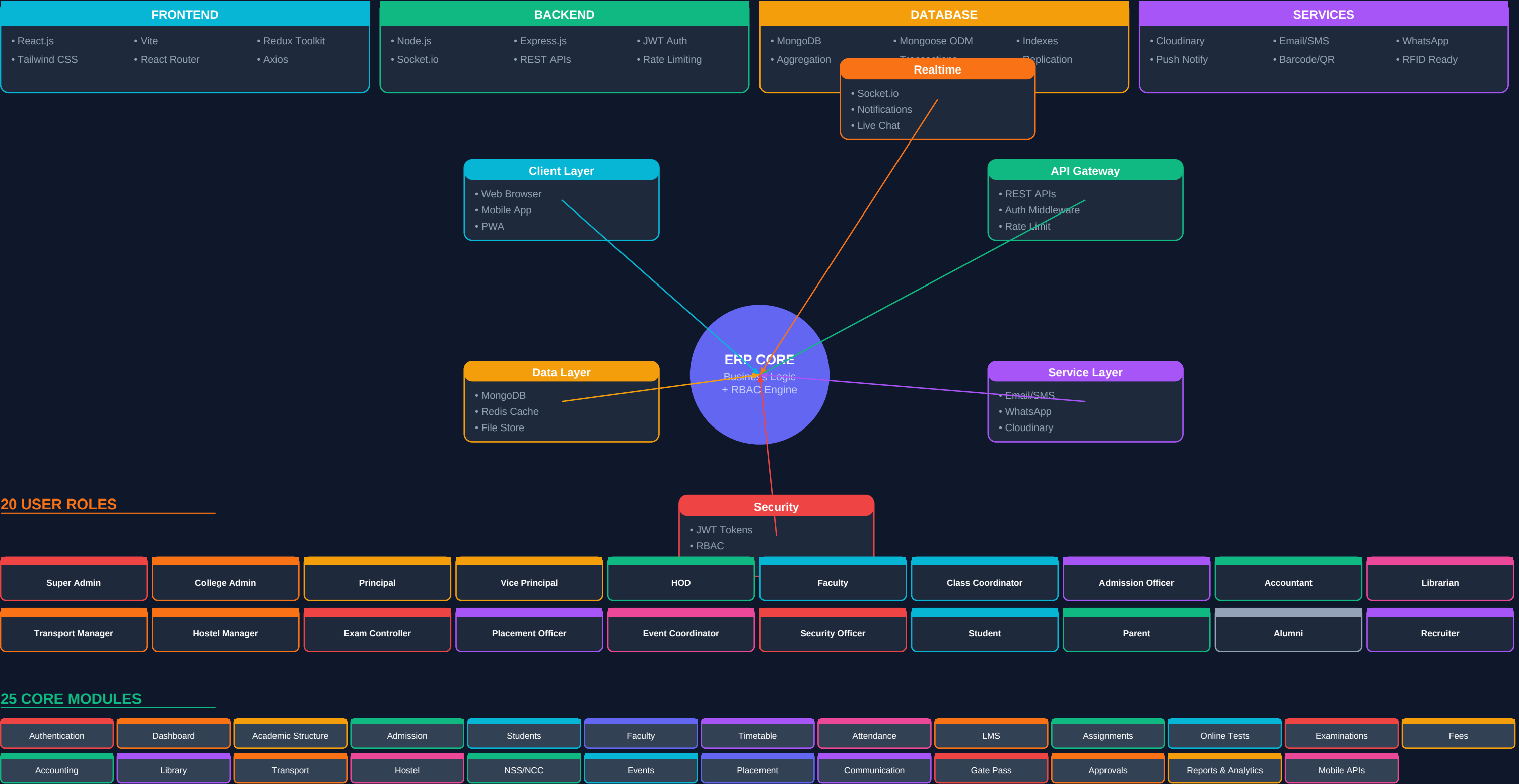


College ERP + LMS + Campus Management System

Complete System Architecture Overview



Authentication & User Access Flow

Login • Register • OTP • JWT • RBAC • Role-Based Dashboards



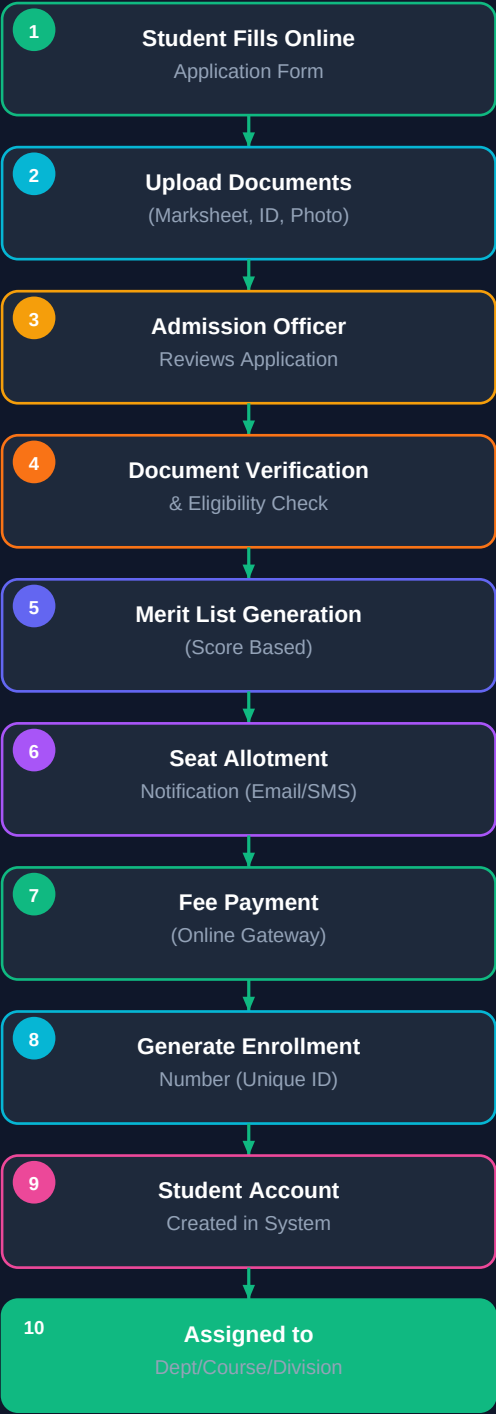
ROLE → DASHBOARD MAPPING

Super Admin	Full System Access: All Colleges, All Modules, Config, Reports
College Admin	College Config, Users, Departments, Finance, All Modules
Principal	Academic Overview, Faculty, Results, Events, Reports
Vice Principal	Academic Reports, Approvals, Timetable, Attendance
HOD	Dept Faculty, Subjects, Timetable, Results, Workload
Faculty	My Timetable, Attendance, LMS Upload, Assignments, Tests
Class Coordinator	Division Attendance, Notices, Student Issues
Admission Officer	Applications, Verification, Merit List, Enrollment
Accountant	Fees, Payments, Ledger, Financial Reports, Receipts
Librarian	Book Catalog, Issue/Return, Fines, Barcode Scan
Transport Manager	Vehicles, Routes, Drivers, Student Allocation
Hostel Manager	Rooms, Allocation, Visitors, Hostel Fees
Exam Controller	Hall Tickets, Seating, Marks Entry, Grades, Results
Placement Officer	Companies, Jobs, Drives, Student Profiles, Stats
Event Coordinator	Create Events, Registrations, Venues, Certificates
Security Officer	Gate Pass, Visitor Log, Student Exit, Approvals
Student	My Courses, Attendance, Assignments, Tests, Fees, Results
Parent	Ward Attendance, Fees, Results, Notices, Transport
Alumni	Network, Job Board, Events, Profile, Referrals
Recruiter	Job Posts, Campus Drive, Student CVs, Interview Slots

Academic Structure & Admission Management Flow

Hierarchy • Online Admission • Merit List • Enrollment • Student Profile

ACADEMIC HIERARCHY



ENROLLMENT ID FORMAT

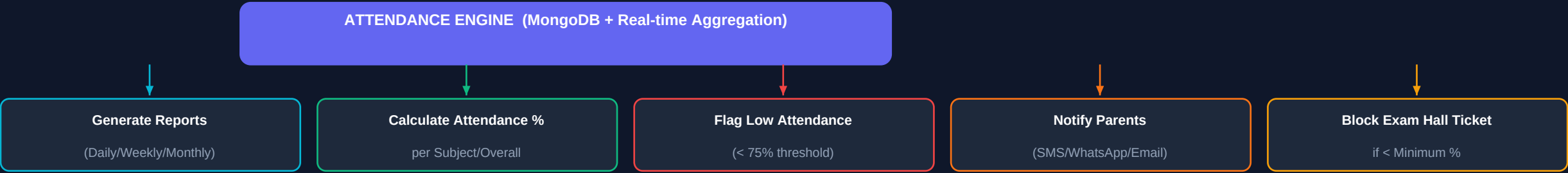
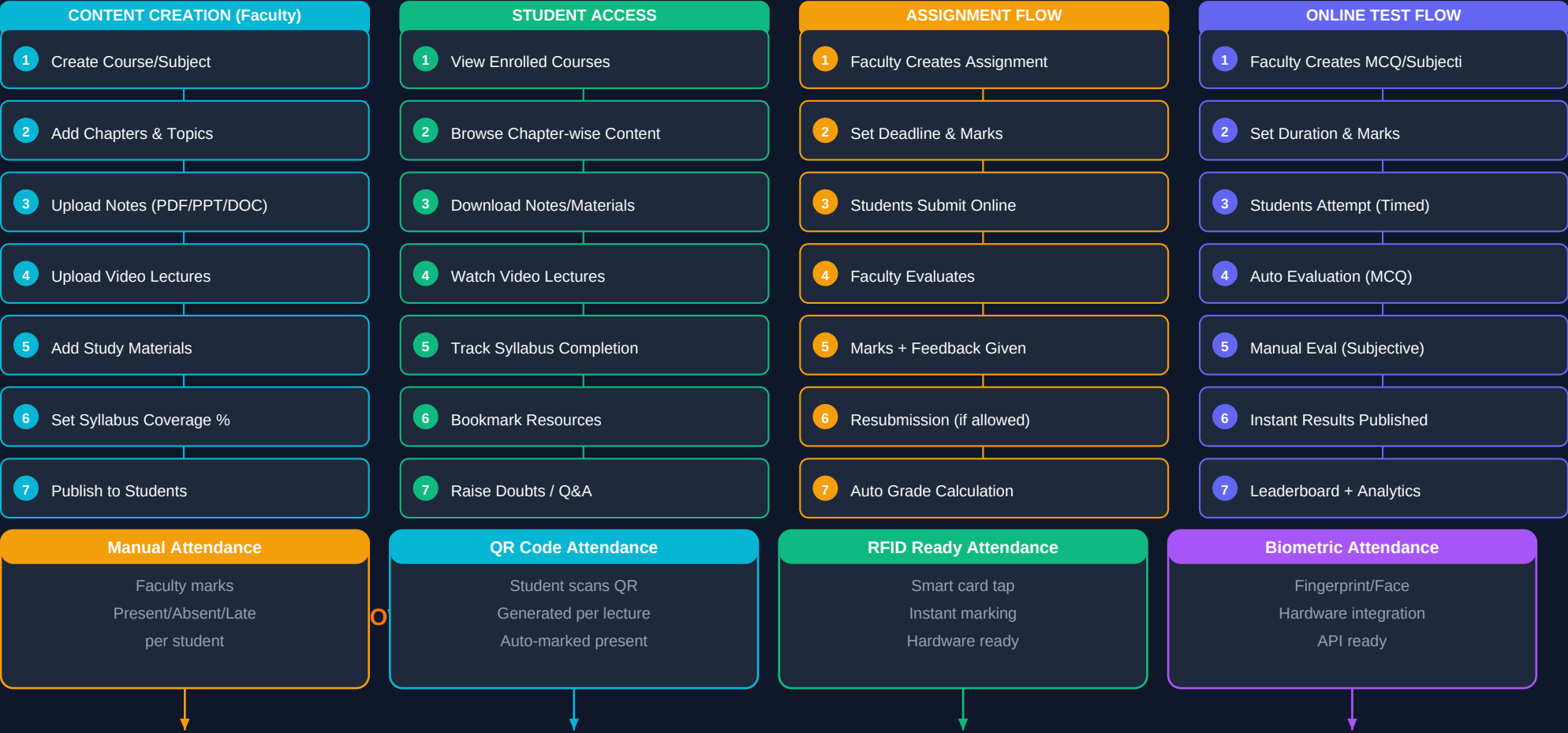


Example: MIT-CS-2024-B1-0047

LMS • Attendance • Timetable Management Flow

Lecture Delivery • Content Upload • Attendance Tracking • Schedule Generation

LMS MODULE FLOW

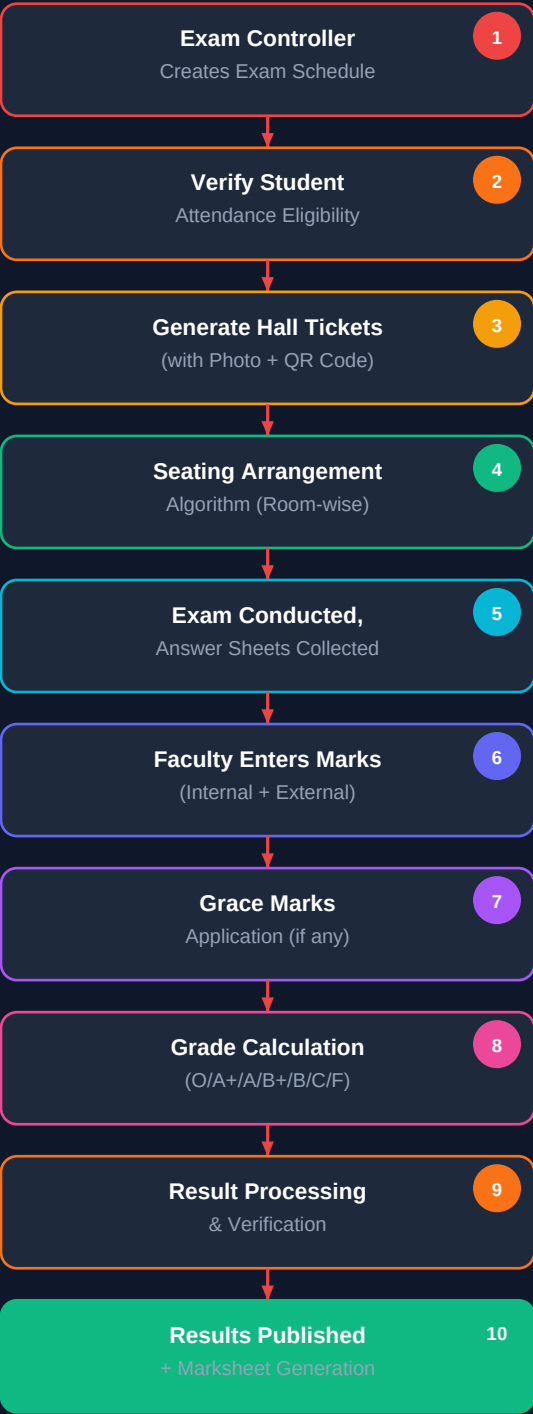


TIMETABLE GENERATION FLOW



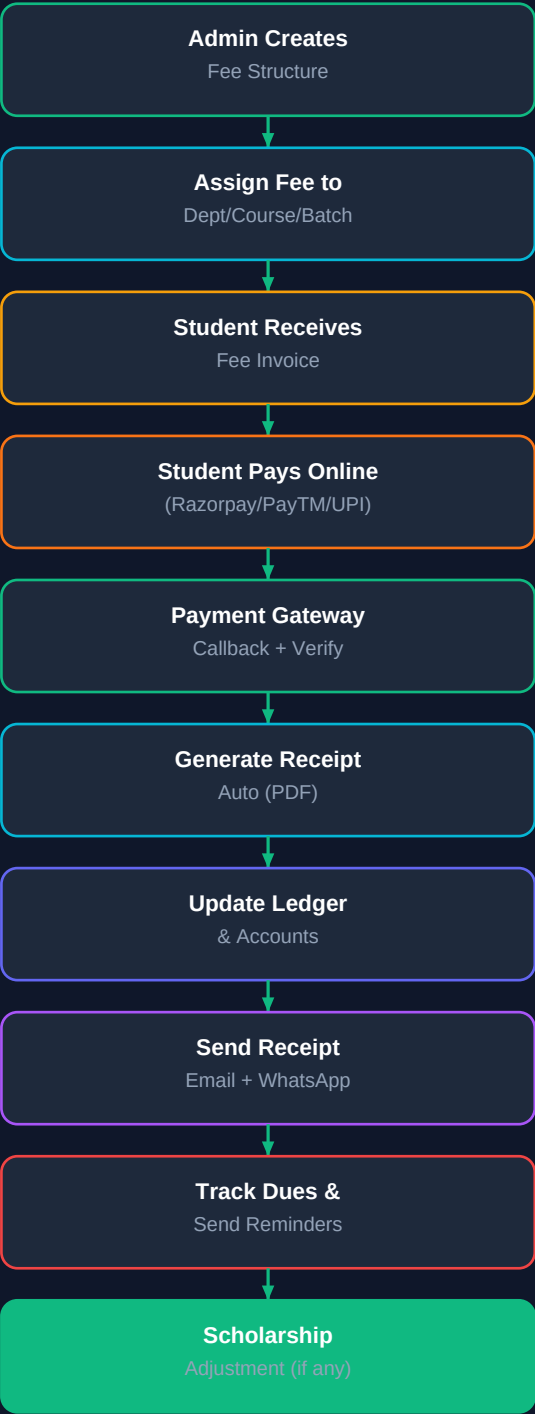
Examination System & Fees / Finance Management Flow

Hall Tickets • Seating • Marks • Grades • Fees Structure • Payments • Accounting



GRADING SCALE

O	91-100% → GP: 10
A+	81-90% → GP: 9
A	71-80% → GP: 8
B+	61-70% → GP: 7
B	51-60% → GP: 6
C	41-50% → GP: 5
P	35-40% → GP: 4
F	0-34% → GP: 0
Ab	Absent% → GP: 0



ACCOUNTING MODULE

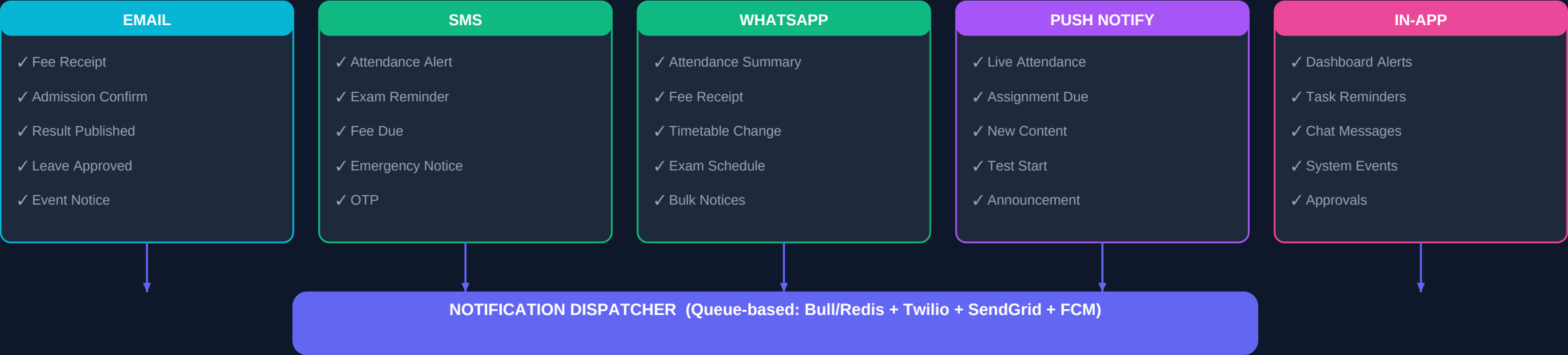
Income Heads	Expense Heads	Reports
Tuition Fees Hostel Fees Transport Fees Library Fines Exam Fees Misc Income	Faculty Salaries Admin Expenses Infrastructure Utilities Events Procurement	Daily Cash Report Monthly P&L Balance Sheet Fee Collection Report Dues Report Audit Trail



Communication • Notifications • Approval Workflows

Multi-Channel Messaging • Leave Management • Purchase Orders • Audit Trail

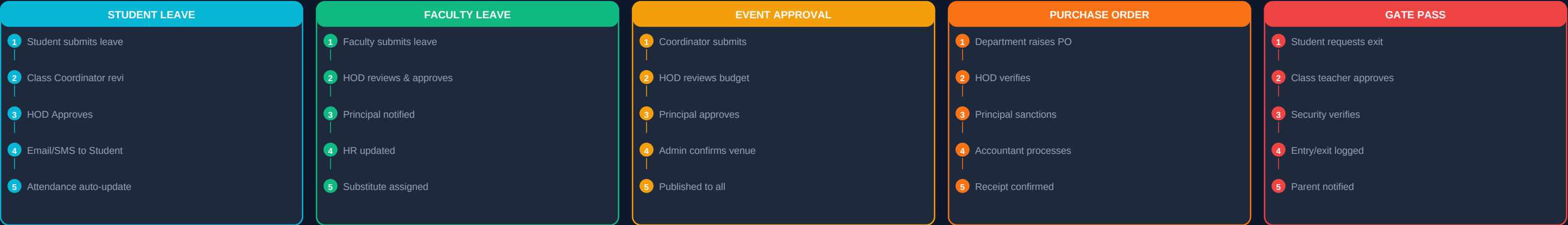
MULTI-CHANNEL NOTIFICATION ENGINE



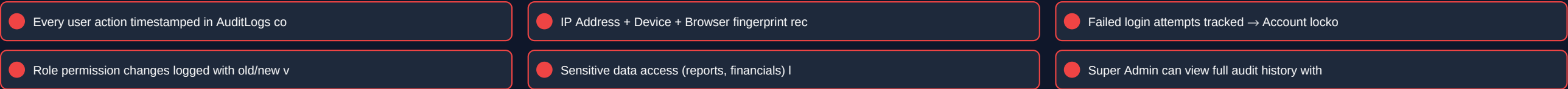
NOTIFICATION TRIGGERS

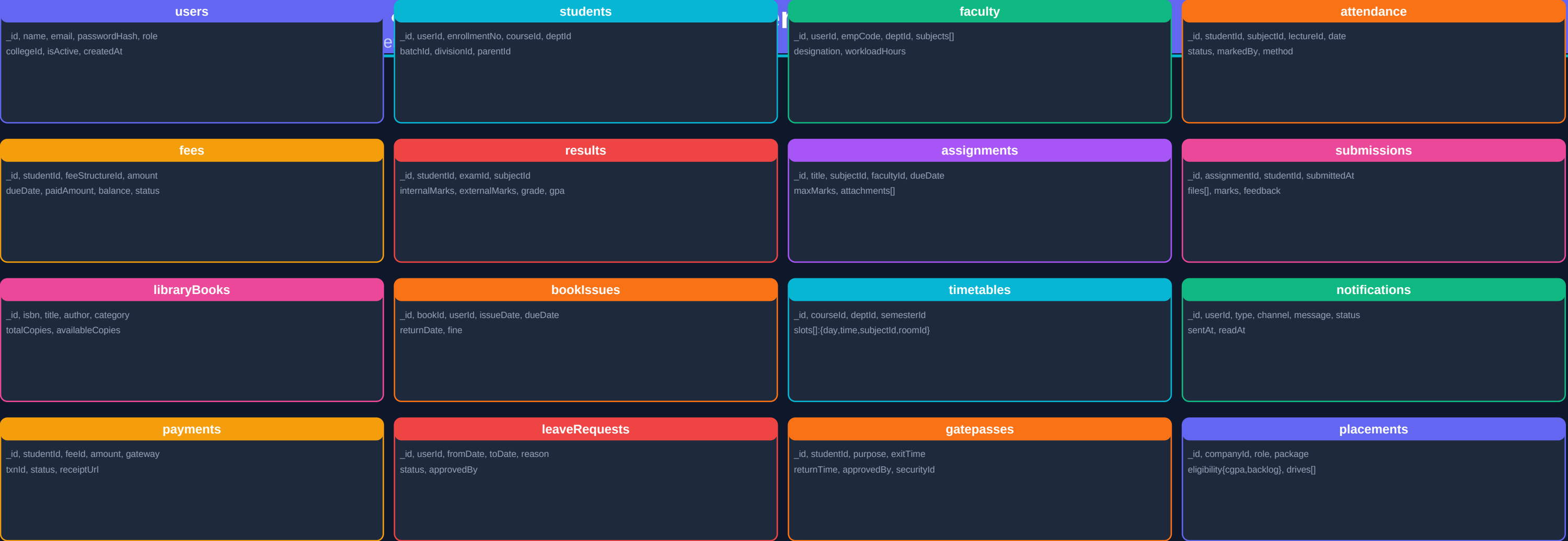


APPROVAL WORKFLOW ENGINE



AUDIT TRAIL & SECURITY LOG

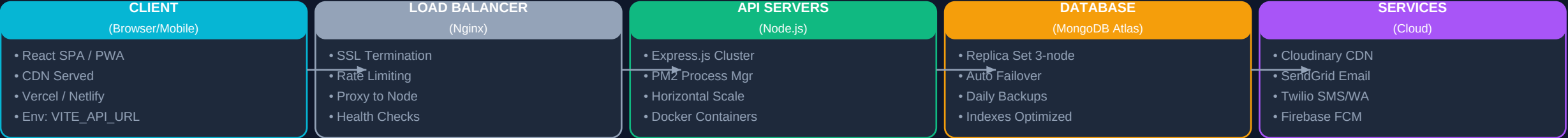




REST API ENDPOINT STRUCTURE

POST /api/auth/login Authenticate user → JWT	POST /api/auth/refresh Refresh access token	GET /api/students List students (RBAC filtered)
POST /api/attendance Mark attendance (bulk)	GET /api/attendance/report Attendance analytics	POST /api/lms/notes Upload study material
GET /api/lms/courses/:id Get course content	POST /api/fees/pay Initiate fee payment	GET /api/results/:studentId Get student results
POST /api/timetable Create/update timetable	GET /api/notifications User notifications	POST /api/leave/request Submit leave request

DEPLOYMENT ARCHITECTURE



DEVELOPMENT PHASES

